

Final Project: Application Proposal:

CYBR 570 : Reverse Engineering Spring 2025

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Title: Analyzing Authentication and Network Security in the FairEmail Android App

Project Description: For my final reverse engineering project, I plan to analyze the FairEmail Android application, an open-source, privacy-first email client available on F-Droid. FairEmail supports OAuth2 authentication and secure IMAP/SMTP communication, making it my fun choice for studying real-world app security.

The goal of this project is to reverse engineer the authentication flow and network communication logic. I will use static analysis tools like JADX and apktool to explore how the app manages login credentials, stores tokens, and interacts with remote APIs. I also plan to perform dynamic analysis using Frida and Radare2 to identify and bypass SSL certificate pinning mechanisms. If time permits, I'd like to simulate API interactions using captured tokens or reconstructed request formats.

Level of Effort/Scope: I expect this to be a moderate-to-high effort project. It will involve analyzing obfuscated code, tracing control flow across Java and native libraries, and dynamically inspecting the app's runtime behavior. I am confident this scope is manageable for myself and I plan to complete this project individually.

Ethical Considerations: FairEmail is fully open-source (GPLv3), and I will only work with a self-built APK using the publicly available source code. All reverse engineering will be done in a controlled, local environment with no real account data or production servers involved. My goal is to understand and document the app's security mechanisms, not to break them. This was the main reason I chose FairEmail since I am legally allowed to reverse engineer the app.

Deliverable Format: I will submit a HOW-TO guide that walks through each stage of my analysis, including tooling, key insights, and any bypasses or observations. The guide will be clearly organized and supported with screenshots, code snippets, and reflections on what I learned through the process.